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Heterogeneous Elastic Network Model of Mycobacterium Smegmatis Porin A

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ABSTRACT

SIGNIFICANCE

INTRODUCTION

The classic representation of proteins has been static, using characterisation techniques such as X-ray crystallography and cryo-electron microscopy (cryo-EM), as these techniques provide an average structure of the proteins. This does not imply that these techniques are not relevant; rather, the fluctuations about that average play a vital role in protein functions. The theoretical approaches for understanding protein structure and function are rapidly developing for studying the dynamics of proteins. Fluctuations play a vital role in many biological phenomena, providing interpretation of protein functions such as binding of ligand and protein, and structural changes due to allosteric effects. Vibrational modes of states affect the free energies of activation and catalytic sites. The electron transfer from donor to acceptor depends strongly on fluctuations. Functions of proteins are understood by investigating the structural fluctuations, dynamics of proteins and the relationship between activity and conformational change (1).

A measurement to quantify motion is obtained from the mean square fluctuations of the atoms from their average positions. These are related to the atomic temperature or Debye-Waller factors, B factors, determined in X-ray diffraction of protein crystals (1). The mean square positional fluctuations assume isotropic and harmonic motion. Normal Mode Analysis (NMA) is a commonly used technique to characterize structural fluctuations in proteins, which assumes that vibrational displacements of the particles from their equilibrium positions are small enough that the potential energy can be approximated as a sum of terms that are quadratic in the displacements. Levy and Karplus (1979) and Levitt *et al*. (1983) employed a complex semi-empirical potential used in molecular dynamics simulations, which comprises bonded (covalent bonds, angles, torsional angle) and non-bonded (Van der Waals, hydrogen bond) interactions (2, 3).

$$V = \sum_{bond} \frac{k_b}{2}(b-b_o)^2 + \sum_{angles} \frac{k_\theta}{2}(\theta-\theta_o)^2 + \sum_{torsion-angles} \frac{k_\phi}{2}(1+cos(n\phi-\delta)+ \\ \sum_{non-bonded-pairs} \frac{A}{r^{12}} - \frac{C}{r^6} + \frac{q_1 q_2}{Dr} + \sum_{hydrogen-bonds} \frac{A'}{r^{12}} - \frac{C'}{r^{10}}$$

Alternative techniques based on NMA with reduced dimensionality for characterizing fluctuations in a protein have been developed. The Elastic Network Model (ENM) adopts a simpler harmonic potential controlled by uniform stiffness as a single parameter. Tirion showed that a single parameter is sufficient to reproduce the experimentally measured fluctuations in proteins at the residue level. Tirion demonstrated that a uniform spring constant,k, with a simplified harmonic potential is sufficient to reproduce the experimentally determined B factors (4).

$$V = \frac{k}{2} \sum_{i,j>i} (r_{ij} - b_{o,ij})^2 \tag{1}$$

Here, we have employed a heterogeneous elastic network model (hENM) developed for characterizing fluctuation and quantifying the correlations at the residue level and domain level fluctuations of a protein, provided it has an average structure. hENM is a tool to build coarse-grained force fields by using the information from all-atom (AA) simulations. The algorithm is similar to the work of Chu *et al.* and Lyman *et al.* (5–7) with some deviations. The model is based on an elastic network model (ENM) using normal mode analysis (NMA) to characterize the fluctuations of the particles. Heterogeneous Elastic Network Model (hENM) was developed with the goal of characterizing collective emotions at multiple resolutions using classical all-atom (AA) simulation trajectories. Fluctuations resulting from anharmonic interactions between atoms in a protein are approximated using a combination of normal modes. Unlike the uniform stiffness of the harmonic potential used in ENM, hENM explicitly accounts for heterogeneity in the stiffness across CG pairs of protein. Molecular simulations with explicit solvent are more representative of thermal fluctuations than crystallographic B-factors (7). We have compared our algorithm with Chu *et al.* and Lyman *et al.* for MspA with the size of the system

METHODS

1 ATOMISTIC SIMULATION OF MSPA EMBEDDED IN A LIPID BILAYER

Mycobacterium smegmatis porin A (Mspa) structure was retrieved from protein data bank (PDB-ID 1UUN) at a resolution 2.5 Å(**?**). It is an octamer (1472 residues) with a unique central channel comprising a vestibule, beta-barrel and a narrow constriction. MspA was inserted in a patch of 171 Å x 171 Å phospholipid bilayer made of 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine molecules (POPC) using CHARMM GUI Input Generator module-Membrane builder (**?**). The complete assembly was solvated in Membrane builder using TIP3P water model with a periodic box of dimension 171 Å x 171 Å x 178 Å consisting of 121329 water residues. The system was neutralized in Membrane builder with a salt concentration of 1M KCl consisting of 2306 K^+ and 2218 Cl^- ions. The whole system consists of 495987 atoms. Atomistic simulation of MspA-bilayer assembly was performed with CHARMM36 forcefield using GROMACS 2024.4 (**?**). The structure was minimized in 5000 steps using the steepest descent algorithm. Equilibration was carried out in different steps using constant volume and temperature (NVT ensemble) and constant pressure and temperature (NPT ensemble). In NVT equilibration, the system was gradually heated for 5ns from 0 K to 298 K and maintained at 298 K for 5ns with restraints on the backbone and side chain of protein and lipids. The restraints on protein and lipids were relaxed in the next two consecutive NVT equilibration steps for 10 ns each at constant volume and temperature 298 K with V-rescale thermostat. The restraints were further relaxed on protein and lipids in the next six consecutive NPT ensembles with a constant pressure of 1 bar and temperature of 298 K using V-rescale thermostat and C-rescale barostat. The last frame was used for the production run with the same coordinates and velocity. Three trajectories were generated from the end of the equilibration step. We have carried out a simulation of MspA in a lipid-bilayer with identical system details and similar protocol for T=310 K.

2 COARSE GRAINING MODEL

A protein can be coarse-grained based on structural, intuition-based or functional-based. In our study, we have coarse-grained MSPA at the multiscale level to their centre of mass. After coarse-graining, we obtain the coarse-grained AA mapped trajectory (CG-AA mapped trajectory) for the all-atom trajectory. We calculate the bond fluctuation of the target CG-AA mapped trajectory between all the CG pairs using the variance:

$$\langle \Delta {r_{ij}}^2 \rangle_{CG-AA} = \langle (r_{ij} - b_{o,ij})^2 \rangle$$

where r_{ij} is the distance between CG pair, and $b_{o,ij}$ is the equilibrium bond length of CG pair. We can use a uniform constant value as an initial guess instead of above mentioned BI method. We have provision for this as well.

3 HETEROGENEOUS ELASTIC NETWORK MODEL (HENM)

We have considered harmonic interaction between the coarse-grained (CG) pair and intend to compute heterogeneous spring constants for each CG pair. The potential form is given by the equation:

$$V = \sum_{i,j>i} \frac{k_{ij}}{2} (r_{ij} - b_{o,ij})^2 \tag{2}$$

where V is total potential, k_{ij} is the spring constant of the CG pair, $b_{o,ij}$ is the equilibrium bond length, r_{ij} is the instantaneous bond length.

The equation of motion is given by Newton's second law of motion. We consider the motion of the system to be harmonic, i.e., a stable configuration at equilibrium. It is assumed that the fluctuations of CG pairs are sufficiently small that the potential can be approximated as a sum of terms that are quadratic in displacement. We can expand the total potential V using the Taylor series about the equilibrium and ignoring the highest-order energy terms (H.O.T).

Substituting in equation **??**, we have,

$$m_i \frac{\partial^2 \epsilon_i}{\partial t^2} = -\sum_{j=1}^{3N} \left\{ \left. \frac{\partial^2 V}{\partial \epsilon_i \partial \epsilon_j} \right|_o \epsilon_j \right\} \tag{3}$$

We can write the above equation as

$$\sqrt{m_i} \frac{\partial^2 \epsilon_i}{\partial t^2} = -\frac{1}{\sqrt{m_i}\sqrt{m_j}} \sum_{j=1}^{3N} \left\{ \left. \frac{\partial^2 V}{\partial \epsilon_i \partial \epsilon_j} \right|_o \epsilon_j \right\} \sqrt{m_j} \tag{4}$$

$$\frac{\partial^2 \tilde{\epsilon}_i}{\partial t^2} = -\sum_{j=1}^{3N} \left\{ \left. \frac{\partial^2 V}{\partial \tilde{\epsilon}_i \partial \tilde{\epsilon}_j} \right|_o \tilde{\epsilon}_j \right\} \tag{5}$$

Here,

$$\tilde{\epsilon}_i = \epsilon_i \sqrt{m_i}$$

is the mass weighted coordinates and

$$\frac{\partial^2 V}{\partial \tilde{\epsilon}_i \partial \tilde{\epsilon}_j} = \frac{1}{\sqrt{m_i}\sqrt{m_j}} \frac{\partial^2 V}{\partial \epsilon_i \partial \epsilon_j}$$

is the mass weighted force field constant Let

$$\tilde{\epsilon} = \tilde{v} sin(\sqrt{\tilde{\lambda}} t)$$

be the solution where, v is the eigenvector and λ is the eigenvalue. We have 3N eigenvalues and 3N eigenvector. This reduces above equation into an Eigenvalue problem,

$$\overline{\overline{\boldsymbol{H}}} \tilde{\mathbf{v}} = \tilde{\lambda} \tilde{\mathbf{v}} \tag{6}$$

If

$$\tilde{\epsilon}_i = \tilde{v} sin(\sqrt{\tilde{\lambda}} t)$$

is the solution, then the linear combination of this is also the solution i.e.,

$$\tilde{\epsilon}_i = \sum_{k=1}^{3N-6} \tilde{v}_k sin(\sqrt{\tilde{\lambda}_k} t) \tag{7}$$

$$\epsilon_i = \frac{\sqrt{2k_B T}}{\sqrt{m_i}} \sum_{k=1}^{3N-6} \frac{\tilde{v}_k sin(\sqrt{\tilde{\lambda}_k} t)}{\sqrt{\tilde{\lambda}_k}} \tag{8}$$

Therefore, the fluctuations obtained using normal mode analysis are given by:

$$\begin{aligned} \langle \Delta r_{ij}{}^2 \rangle_{NMA} = k_B T \sum_{k=1}^{3N-6} \Bigg\{ & \frac{1}{\sqrt{\tilde{\lambda}_k}} \left(\frac{x_i - x_j}{dr_{ij,em}} \right) \left(\frac{\tilde{v}_{k,x_i}}{\sqrt{m_i}} - \frac{\tilde{v}_{k,x_j}}{\sqrt{m_j}} \right) + \frac{1}{\sqrt{\tilde{\lambda}_k}} \left(\frac{y_i - y_j}{dr_{ij,em}} \right) \left(\frac{\tilde{v}_{k,y_i}}{\sqrt{m_i}} - \frac{\tilde{v}_{k,y_j}}{\sqrt{m_j}} \right) \\ & + \frac{1}{\sqrt{\tilde{\lambda}_k}} \left(\frac{z_i - z_j}{dr_{ij,em}} \right) \left(\frac{\tilde{v}_{k,z_i}}{\sqrt{m_i}} - \frac{\tilde{v}_{k,z_j}}{\sqrt{m_j}} \right) \Bigg\}^2 \end{aligned}$$

4 DETERMINATION OF FORCE-FIELD

Then, we minimize the error $(\langle \Delta r_{ij}{}^2 \rangle^n_{NMA} - \langle \Delta r_{ij}{}^2 \rangle_{CG-AA})$ between the fluctuations computed using normal mode analysis and target CG-AA mapped trajectory using Newton Raphson method.

$$k^{n+1}_{ij} = k^n_{ij} - \alpha \frac{f\left(k^n_{ij}\right)}{f\left(\left(k^n_{ij}\right) + \beta\right)) - f\left(k^n_{ij}\right)/\beta} \tag{9}$$

where $f\left(k^n_{ij}\right) = (\langle \Delta r_{ij}{}^2 \rangle^n_{NMA} - \langle \Delta r_{ij}{}^2 \rangle_{CG-AA})^2$

Here, n denotes the iteration, and parameters α and β control the increment in spring constant and the size of the steps, respectively.

After minimizing the error, the obtained spring constants are used further for coarse-grained simulation. The goal is to map the bond fluctuation and positional fluctuation as that of the target CG-AA mapped trajectory.

Other methods for convergence used by Lyman *et al.* (7) and Chu *et al.* (5) are:

Lyman *et al.* (7):

$$\frac{1}{k^{n+1}_{ij}} = \frac{1}{k^n_{ij}} - \alpha(\langle \Delta r_{ij}{}^2 \rangle^n_{NMA} - \langle \Delta r_{ij}{}^2 \rangle_{CG-AA}) \tag{10}$$

Chu *et al.* (5):

$$k^{n+1}_{ij} = k^n_{ij} + \alpha(\langle \Delta r_{ij}{}^2 \rangle^n_{NMA} - \langle \Delta r_{ij}{}^2 \rangle_{CG-AA}) \tag{11}$$

5 RESULTS AND DISCUSSION

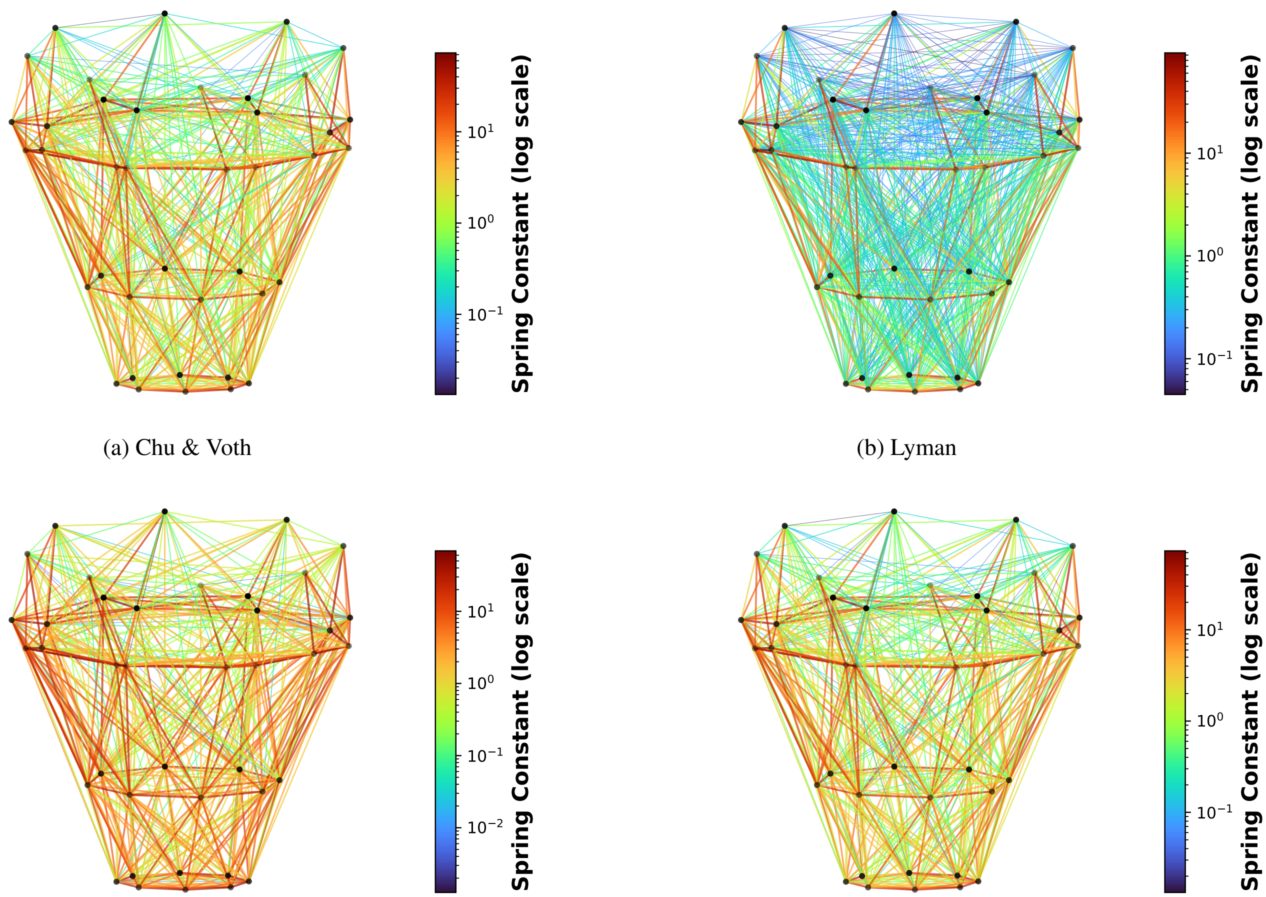
5.1 40 Coarse grained model of MspA

We have constructed a 40 CG model of MspA with each CG site connected to all the other CG sites, i.e., 780 bonds. There are $\frac{N(N-1)}{2}$ number of CG sites where N is the number of CG sites if each bond is connected to all the other sites. We have not used any cutoff in this model. A network model consists of nodes and edges. The nodes of the network represent the CG particles, and the edges represent the contacts (bonds) among the nodes (particles). The links (1,2) and (2,1) are expressed as the same. We have used the same temperature as in the MD simulation, i.e., T = 298 K. We have a 900 ns MspA CG-mapped trajectory with an aligned trajectory. We have estimated the initial spring stiffness using the Boltzmann Inversion method, unlike the uniform arbitrary value used in the Lyman *et al.* and Chu *et al.* method to reduce computational time. All computations were performed on a system equipped with an AMD Ryzen 7 5700G CPU (4.0 GHz), 27 GB RAM, running Ubuntu 22.04.

We have taken the parameters α value equal to 0.1, β equal to 0.1 and the tolerance for the sum of square of error (SOE) was kept to be 5 for all the methods, i.e., Lyman method, Chu & Voth method, HENM and HENM Newton Raphson (NR) method. The value was fixed at 5, as the SOE converges to a constant for these parameters and does not fall below this tolerance. We have estimated the force constants using a mass of 1 gm/mol of the CG sites. Considering the number of bonds (nb) and number of iterations (itr), the algorithm performs (nb x itr) steps. In the HENM method, perturb all the spring constants together in one iteration and estimate the spring constants. We have implemented that if any bonds become a negative value, they are set to zero in each iteration, as in the Lyman method. We have compared the NR and the HENM method with the Lyman and Chu & Voth method.

$$SOE = \sum_{nb} (\langle \Delta r_{ij}{}^2 \rangle^n_{NMA} - \langle \Delta r_{ij}{}^2 \rangle_{CG-AA})^2 \tag{12}$$

The convergence behavior of the different parameterization methods was evaluated by monitoring the sum of squared errors (SOE) during the iterative optimization procedure (Fig. 2). The SOE measures the discrepancy between the coarse-grained elastic network model predictions and the reference data. As the iterations proceed, the SOE decreases monotonically, indicating progressive improvement of the spring constant parameters. Comparison of the convergence curves reveals significant differences in the efficiency of the methods. In particular, the proposed NR approach exhibits the fastest decrease in SOE, reaching a cutoff of 1 within a small number of iterations, with lesser computation time of around 1.8 minutes, whereas the Chu–Voth and Lyman approaches require substantially more iterations and more computation time of around 8 and 5 minutes, respectively, to achieve



(a) Chu & Voth

(b) Lyman

(c) HENM

(d) HENM-NR

Figure 1: Comparison of spring networks obtained using four different methods for MspA protein for a 40 CG site system .The Spring Constants shown as colored lines. The stiffness of each spring is mapped to color on a logarithmic scale which may vary based on the range of values, with units of kJ mol^{-1} Å^{-2}, with larger spring constants represented by warmer colors and smaller constants by cooler colors. The resulting network highlights that the our new method forms a similar to network to Chu & Voth and Lyman and we achieve this with less number of bonds between CG sites.

comparable accuracy. Even though the processing time seems comparable for all the four methods but for a bigger system like for 184 CG sites the processing time effect more profound.

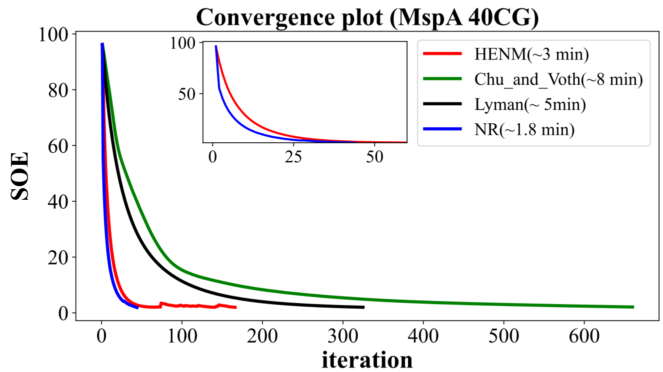


Figure 2: Sum of error computed for all 4 methods using Eq(12) here we can clearly see that both Chu&Voth and Lyman take much more iterations than our proposed methods which take significantly less iterations to reach the convergence cutoff

On the same note as Figure 1 here we can see the effect of equilibrium bond length on the spring constant value as scatter of spring constants (kJ mol^{-1} Å^{-2}) are plotted against the equilibrium bond length (Å), for all the four methods where the equilibrium bond length increases as we move along the x axis and the bond stiffness decreases. This behavior is consistent with the findings of Chu *et al.* and Lyman *et al.* (5–7) showing that shorter bonds show higher stiffness and longer bonds show lesser stiffness. The Lyman method shows a broader distribution with higher stiffness values at short distances, reflecting its use of a dense, fully connected network. In contrast, the Chu & Voth method produces a sparser network with comparatively lower stiffness values. The proposed NR method captures a similar trend while maintaining a reduced network size, yielding stiffness distributions comparable to existing methods.

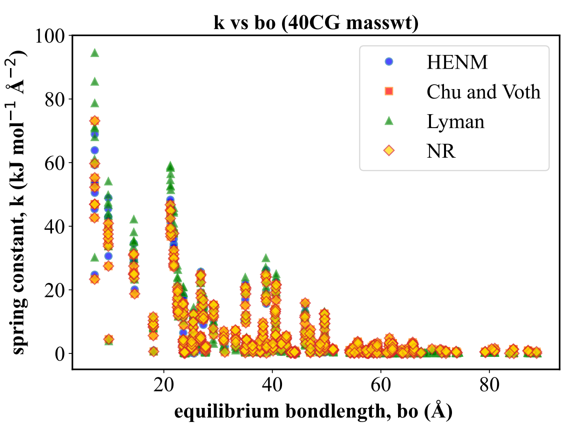


Figure 3: This indicates Spring Constant (kJ mol^{-1} Å^{-2})

The SOE convergence alone is not sufficient to verify the correctness of the proposed method. Therefore, coarse-grained (CG) simulations were performed and compared with all-atom molecular dynamics (MD) simulations. The spring constants obtained at the end of the SOE convergence were used to construct the CG model, considering only those values for which k>0. Bond fluctuations were calculated for all bonded pairs, and the positional root-mean-square fluctuations (RMSF) were computed as a function of the residue index. The reference data were generated using an MD simulation of 900 ns. In contrast, the CG simulations were performed using Langevin dynamics with a time step of 1 fs for a total simulation time of 300 ns at a temperature of 298 K. The damping coefficient was set to 1, and harmonic bonds were constructed using the obtained spring constants. The mass of each CG site was taken as unity. In Figures 4–7, the right panels show the CG bond fluctuations compared with the CG-mapped MD simulation results for all four methods, while the left panels show the per-residue positional fluctuations. These results demonstrate that the Lyman method employs a full network and converges more slowly before achieving comparable results. The Chu & Voth approach uses a smaller network but requires a significantly larger number of iterations to converge. In contrast, the proposed method uses a network size comparable to that of Chu & Voth, but converges in substantially fewer iterations while producing results consistent with the MD simulations.

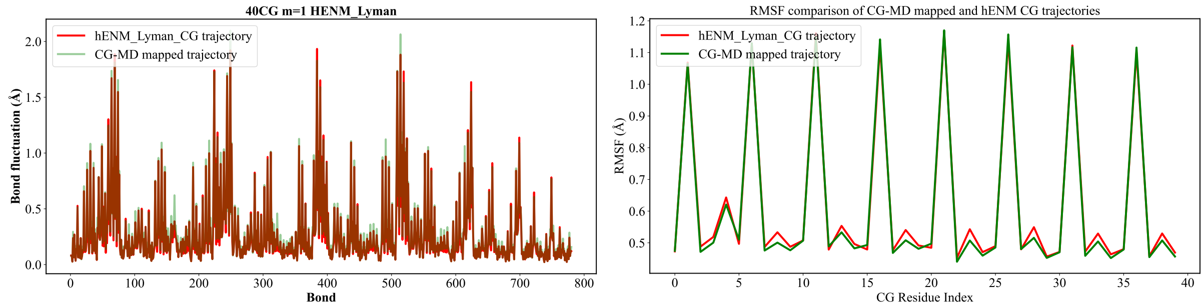


Figure 4: (Right) per bond index Bond fluctuations calculated with Lyman method is mapped with CG mapped MD simulation, (left) per residue index root mean square fluctuations are mapped with CG mapped MD simulation

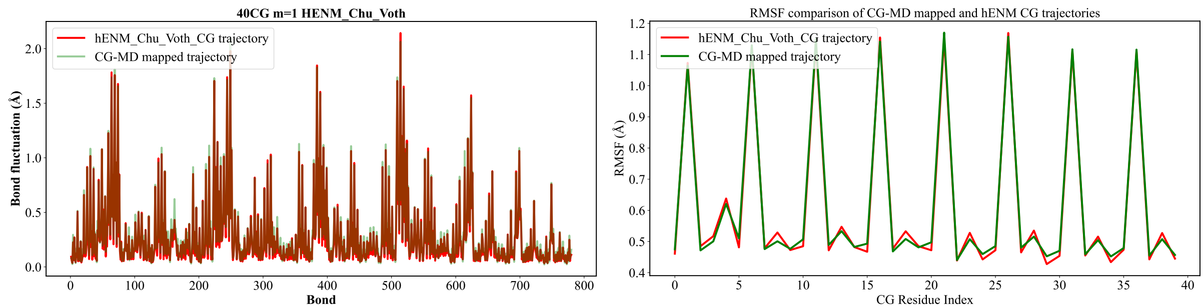


Figure 5: (Right) per bond index Bond fluctuations calculated with Chu&Voth method is mapped with CG mapped MD simulation, (left) per residue index root mean square fluctuations are mapped with CG mapped MD simulation using Chu&Voth method

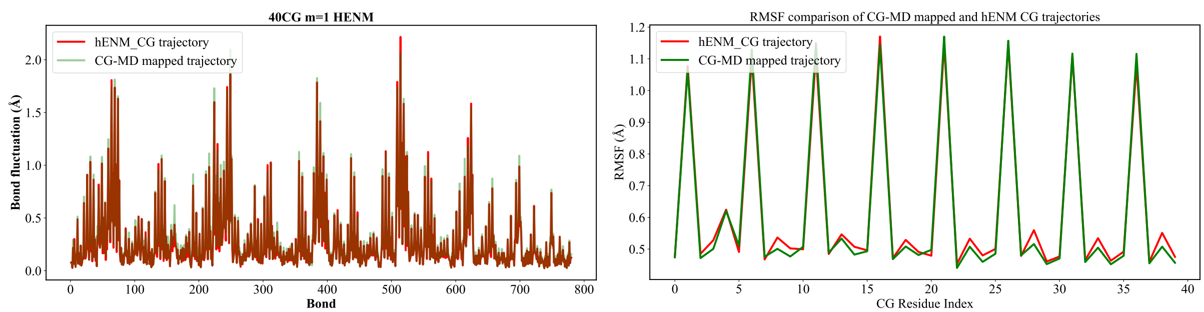


Figure 6: (Right) per bond index Bond fluctuations calculated with henm method is mapped with CG mapped MD simulation, (left) per residue index root mean square fluctuations are mapped with CG mapped MD simulation using henm method

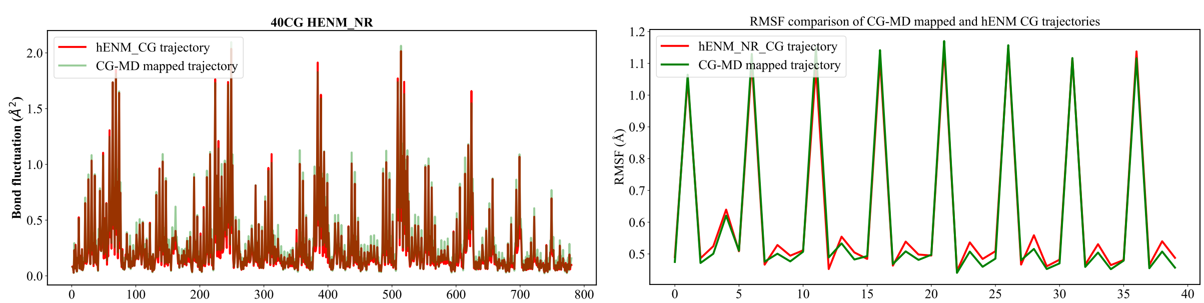
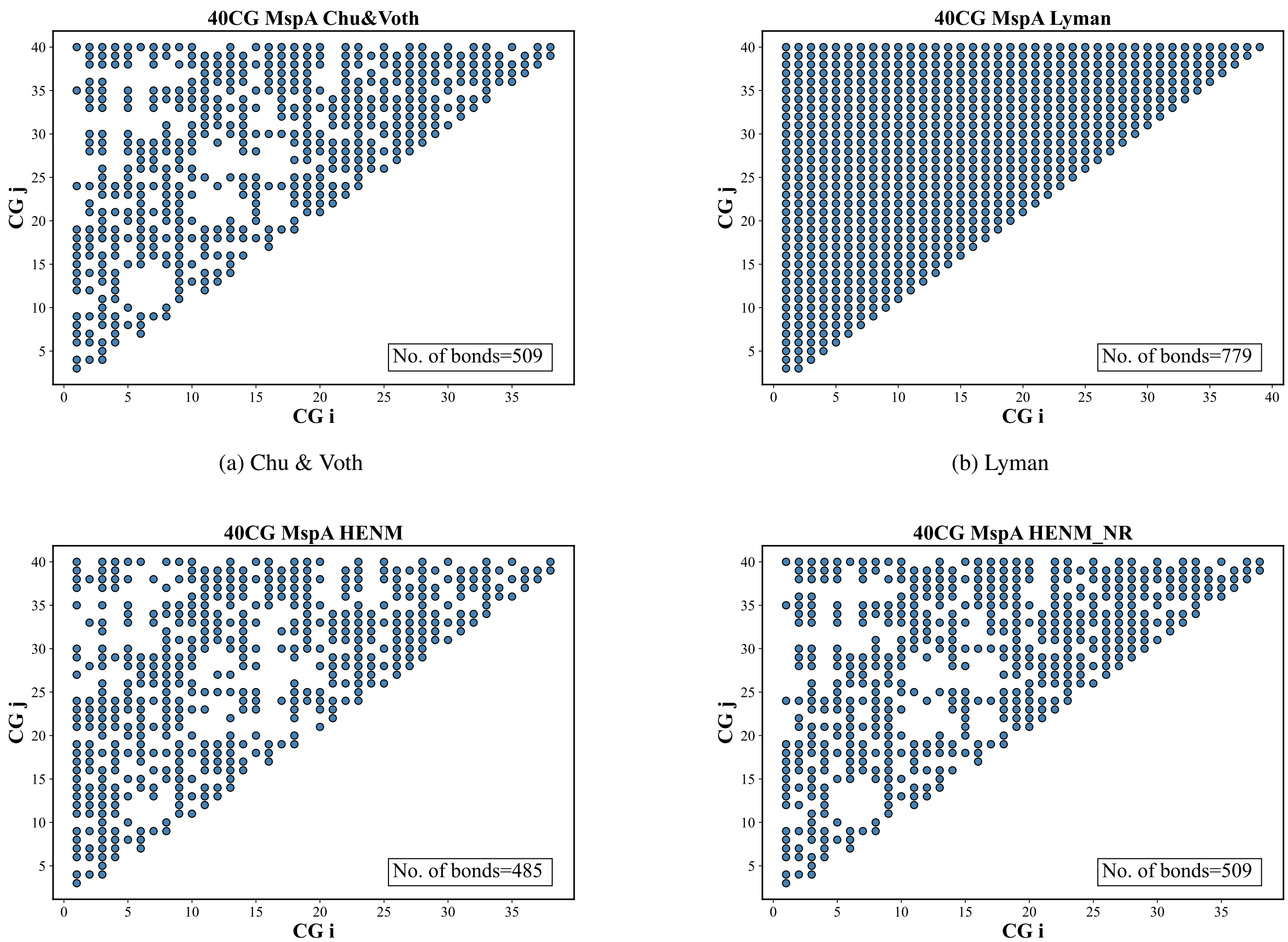


Figure 7: (Right) per bond index Bond fluctuations calculated with NR method is mapped with CG mapped MD simulation, (left) per residue index root mean square fluctuations are mapped with CG mapped MD simulation using NR method

To better understand the CG network, we look into Figure 8 which illustrates the coarse-grained (CG) network obtained after SOE convergence. A blue marker between CG sites i and j indicates that the corresponding spring constant is greater than zero (k>0). The absence of a marker (empty regions) denotes that the spring constant between those CG sites was set to zero at the end of the convergence, as the computed values were negative (k<0). It is very evident that Lyman uses uses a full blown network and Chu & Voth uses a smaller network but both of these methods take a lot of time to converge notable in our proposed methods uses a smaller network comparable to Chu & Voth and takes less number of iteration steps to converge



(a) Chu & Voth

(b) Lyman

(c) Henm

(d) Henm NR

Figure 8: Comparison of bond networks obtained using four different methods: (a) Chu & Voth, (b) Lyman, (c) HENM, and (d) HENM-NR. Each point represents a bond between CG site i and CG site j included in the CG network. The Lyman method generates the densest and nearly full network, whereas Chu & Voth, HENM, and HENM-NR produce sparser networks. Among the reduced-network approaches, HENM-NR retains a connectivity pattern comparable to the other methods while using fewer effective bonds.

With N being small, all methods are equally effective, but as N increases, the NR method becomes computationally expensive.

5.2 184 Coarse grained model of MspA

CONCLUSION

AUTHOR CONTRIBUTIONS

Author1 designed the research. Author2 carried out all simulations and analyzed the data. Author1 and Author2 wrote the article.

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SUPPLEMENTARY MATERIAL

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